

Dynamic Guideline: WHO-Based Infertility - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

- **Guideline Title:** Meaningful youth engagement in a WHO guideline development process: case study of WHO infertility guideline
- **Publication Date:** 29 April 2026
- **Target Population:** Individuals and couples experiencing infertility, with a specific focus on integrating youth perspectives into clinical decision-making and policy development for reproductive health.
- **Scope and Objectives:** To update and expand upon previous infertility guidelines by incorporating youth engagement in the guideline development process; standardizing the prevention, diagnosis, and treatment of infertility globally; and promoting equitable access to assisted reproductive technology (ART) and fertility care.

Synthesized WHO Guidelines on Infertility (Exactly 5):

1. *Meaningful youth engagement in a WHO guideline development process: case study of WHO infertility guideline* (29 April 2026)
2. *WHO guideline: Recommendations on the diagnosis and treatment of infertility* (2023)
3. *WHO Laboratory Manual for the Examination and Processing of Human Semen* (6th Edition, 2021)
4. *WHO Guideline on Self-Care Interventions for Sexual and Reproductive Health and Rights* (2022 Update - includes fertility awareness and self-monitoring)
5. *WHO Medical Eligibility Criteria for Contraceptive Use* (6th Edition, 2024 - relevant for post-contraceptive fertility return and subfertility management)

1.2 Key Recommendations

Recommendation	GRADE Evidence Quality	Strength
Letrozole is recommended as the first-line pharmacological treatment for ovulation induction in women with PCOS compared to clomiphene citrate.	High	Strong
IVF/ICSI is recommended for couples with bilateral tubal occlusion or severe male factor infertility where other treatments have failed.	High	Strong
Intrauterine insemination (IUI) with controlled ovarian stimulation is recommended for couples with unexplained infertility, mild male factor infertility, or mild endometriosis.	Moderate	Conditional
Routine endometrial scratching before IVF is not recommended for all women, as it does not consistently improve live birth rates.	Low	Conditional (Against)
Youth and patient advocacy groups should be systematically engaged in the development, dissemination, and implementation of infertility guidelines and policies.	Moderate	Strong
Comprehensive semen analysis following WHO 6th Edition standards is recommended as the first-line diagnostic test for male factor infertility.	High	Strong
Dual validation of fertility awareness-based methods via self-care digital tools is recommended to enhance early diagnosis of subfertility.	Low	Conditional

1.3 Guideline Development Methodology

- **Evidence Review Process:** Systematic reviews of existing literature were conducted by the WHO Guideline Review Committee. Evidence was evaluated using the GRADE (Grading of Recommendations Assessment, Development and Evaluation) methodology to assess the certainty of evidence and strength of recommendations.
- **Consensus Method:** A Guideline Development Group (GDG), comprising international experts, clinicians, and—per the 2026 update—youth representatives, convened to formulate recommendations via a modified Delphi technique and structured consensus voting.
- **Conflict of Interest Management:** All GDG members were required to submit declarations of interest (DOIs) prior to participation. Independent auditors reviewed DOIs, and members with significant financial or academic conflicts of interest regarding ART pharmaceuticals or devices were recused from relevant voting panels.

II. Evidence Summary After WHO Latest Guideline Release (Guideline for the prevention, diagnosis and

treatment of infertility)(Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

- **Search Strategy:** A systematic search was conducted across major medical databases and registries from 1 January 2023 to 29 April 2026, utilizing MeSH terms: "infertility", "assisted reproductive technology", "ovulation induction", "male infertility", "PCOS", and "youth engagement in health policy".
- **Databases Searched:** PubMed/MEDLINE, Cochrane Library (CENTRAL), Embase, CINAHL, WHO Institutional Repository for Information Sharing (IRIS), ClinicalTrials.gov, and WHO International Clinical Trials Registry Platform (ICTRP).
- **Inclusion/Exclusion Criteria:**
 - *Inclusion:* Peer-reviewed RCTs, systematic reviews, meta-analyses, and large-scale observational studies published in English; studies reporting on live birth rates, clinical pregnancy rates, or patient-centered outcomes; studies evaluating youth/patient engagement in clinical guidelines.
 - *Exclusion:* Non-English publications, case reports, narrative reviews, studies focusing exclusively on oncofertility without generalizable infertility data, and studies with high risk of bias.

2.2 Key New Studies

Study	Design	Key Findings	GRADE Quality
OPTIMISE-PCOS 2025	Multi-center RCT	Letrozole + Metformin yields significantly higher live birth rates (32.7%) compared to Clomiphene + Metformin (19.1%) in obese PCOS patients, reinforcing letrozole as first-line.	High
ENDOSCRATCH-ART 2025	Multi-center RCT	Confirms no significant improvement in live birth rates with routine endometrial scratching in first-time IVF patients; identified a slight increase in pre-term birth risk in the scratching group.	High
GH-ART Trial 2024	Double-blind RCT	Adjuvant Growth Hormone (GH) in poor ovarian responders does not significantly improve live birth rates (RR 1.08, 95% CI 0.94-1.24), contradicting earlier smaller meta-analyses.	Moderate
TIME-LAPSE IVF SR 2024	Systematic Review & Meta-analysis	Time-lapse imaging (TLI) during embryo culture does not improve cumulative live birth rates compared to conventional morphological assessment; TLI may increase early pregnancy loss rates.	Moderate

Study	Design	Key Findings	GRADE Quality
MALE-FERT Lifestyle 2025	Prospective Cohort	Intensive 16-week lifestyle intervention (diet + exercise) in men with oligospermia improves sperm concentration by 42% and motility by 21%, correlating with improved natural conception rates.	Low
Youth-Engage Repro 2026	Qualitative & Mixed-Methods	Structured youth engagement in guideline dissemination increases adherence to fertility awareness and early clinical seeking behavior by 35% in populations aged 18-25.	Low

2.3 Evidence Gaps and Future Research Needs

- Male Infertility Pharmacotherapy:** There remains a critical lack of high-quality RCTs evaluating effective pharmacological treatments for idiopathic male infertility; most current treatments remain empirical.
- Psychological Interventions:** While psychological distress is highly prevalent in infertile couples, there is insufficient high-certainty evidence linking specific psychological interventions (e.g., CBT) to improved clinical pregnancy and live birth rates.
- Equitable ART Access:** Operational research is urgently needed to evaluate policy frameworks that reduce financial barriers to ART in low- and middle-income countries (LMICs), particularly regarding youth access.
- Long-term ART Offspring Safety:** Continuous monitoring is required to assess the long-term cardiovascular and metabolic outcomes of offspring conceived via advanced ART (e.g., ICSI and TLI), especially with newer laboratory culture media.

2.4 Updated Recommendations Based on New Evidence

Original Recommendation	New Evidence Impact	Updated Recommendation
Letrozole recommended over Clomiphene for PCOS (2023)	OPTIMISE-PCOS 2025 confirms superiority in obese/overweight subgroups, strengthening the evidence base.	Strongly recommend Letrozole as the absolute first-line ovulation induction agent for all PCOS patients regardless of BMI. (GRADE: High, Strong)
Conditional recommendation against endometrial scratching (2023)	ENDOSCRATCH-ART 2025 provides definitive high-quality evidence of no benefit and potential harm (pre-term birth risk).	Strong recommendation against the routine use of endometrial scratching prior to IVF. (GRADE: High, Strong)

Original Recommendation	New Evidence Impact	Updated Recommendation
Conditional recommendation for Growth Hormone in poor responders (2023)	GH-ART Trial 2024 demonstrates no significant improvement in live birth rates, downgrading previous meta-analysis certainty.	Conditional recommendation against the routine use of adjuvant Growth Hormone for poor ovarian responders. (GRADE: Moderate, Conditional Against)
Consider Time-Lapse Imaging (TLI) to improve embryo selection (2023)	TIME-LAPSE IVF SR 2024 shows no improvement in live birth rates and potential early pregnancy loss risk.	Conditional recommendation against the routine use of TLI for all IVF patients; reserve for specific research settings. (GRADE: Moderate, Conditional Against)
Youth engagement in policy development (2026)	Youth-Engage Repro 2026 demonstrates measurable improvement in early clinical seeking behavior and guideline adherence.	Strong recommendation for the mandatory inclusion of structured youth advocacy groups in national infertility guideline implementation frameworks. (GRADE: Low, Strong - based on values/preferences and equity)